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Appendix

A RICCI CURVATURE AND CONVERGENCE OF GRADIENT DESCENT

We establish a theoretical link between the ORC of the attention map and the convergence of gradient descent on transformers. As we discuss in Section 4.1, under a set of standard assumptions, this connection follows from the theory of non-convex optimization of over-parameterized neural networks, the eigenvalue spectrum of transformers, and the ORC bound on the graph spectrum.

Recall from Section 4.1 that given the attention matrix A in a transformer $F(\cdot)$ with L identical attention blocks, we can write the output of the transformer as

$$F(\mathbf{x}) = (I + H \otimes A)^L \mathbf{x}, \tag{9}$$

where $H \coloneqq W_p^T W_V^T$. Our arguments implied from Lemma 4.2 state that the minimum ORC of A is linked to the probability of exponential convergence of gradient descent on F, denoted p. In particular, Lemma 4.2 establishes a correlation between p and the the minimum eigenvalue of A, and we show that this eigenvalue is bounded by a monotonic function of the minimum ORC of A. In this appendix, we specify the assumptions leading to this finding and provide a proof.

A.1 ASSUMPTIONS

The majority of our assumptions follow the setup in Dovonon et al. (2024). We however introduce two other assumptions that we discuss further in this section. All assumptions are listed below.

A0. The transformer uses a self-attention mechanism, and the keys and queries share a projection matrix, i.e. $K = Q$.

A1. A is positive and invertible, and all its eigenvalues are real.

A2. Eigenvalues of H are positive and bounded, i.e., $0 < {\lambda^H}_1 < \ldots < {\lambda^H}_d < \infty$

A3. ${\lambda^H}_d < |{\lambda^A}_1|^{-1}$ where ${\lambda^A}_1$ is the smallest eigenvalue of A.

A4. Let $M \coloneqq I + H \otimes A$, and let θ be the vector of the weights in F, consisting of the entries of the weight matrices W_Q, W_K, W_V, and W_p. Then $\nabla_\theta M$ and M are simultanously diagonalizable by an orthogonal matrix P, i.e., there exists an orthogonal matrix P and diagonal matrices D and $\tilde{D}$, such that $M = P^{-1}DP$ and $\nabla_\theta M = P^{-1}\tilde{D}P$. Furthermore, we assume $\tilde{D}$ is approximately constant with respect to D.

A5. The minimum eigenvalue of A is negative.

Note that assumptions **A1** and **A2** follow the setup by Dovonon et al. (2024), and imply

$$-1 < {\lambda^A}_1 \leq \ldots \leq {\lambda^A}_n = 1. \tag{10}$$

A.2 PROOF

Considering that $F(\mathbf{x}) = (I + H \otimes A)^L \mathbf{x}$, we can write $F(\mathbf{x})$ as a linear operator $F(\mathbf{x}) = F\mathbf{x}$ where $F = (I + H \otimes A)^L$. Hence, the eigenvalues of F are $\left(1 + {\lambda^A}_i {\lambda^H}_j\right)^L$.
Let $\varphi \coloneqq (\nabla_\theta F)(\nabla_\theta F)^T$ be the tangent kernel of F. We can write $\varphi(\mathbf{x}) = \nabla_\theta M^T B B^T \nabla_\theta M$ where $M \coloneqq I + H \otimes A$ and $B \coloneqq M^{L-1}$. By assumption **A4**, we can write $\varphi(\mathbf{x}) = P^{-1}\tilde{D}D^{2L-2}\tilde{D}P$. Since diagonal matrices are commutative, it follows that the eigenvalues of φ are of the form

$$\lambda_m^\varphi = \eta_m^2 \left(1 + \lambda_i^A \lambda_j^H\right)^{2L-2}, \tag{11}$$

where η_m is a diagonal entry of $\tilde{D}$. On the other hand, it follows from **A2**, **A4**, and **A5** that the minimum eigenvalue of φ is $\lambda_1^\varphi = \eta_m(1 + {\lambda^A}_1 {\lambda^H}_d)^{2L-2}$ for some m, since by **A2** and **A4** we know $0 < 1 + {\lambda^A}_1 {\lambda^H}_j < 1$ and achieves its minimum values for $j = d$. Given μ, let p be the probability that F is $\mu-$PL, i.e., the PL theorem holds as described in Section 4.1. From Theorem 4.1, we have

$$p = \mathbb{P}\left[\lambda_1^\varphi \geq \mu\right]. \tag{12}$$

Note that we can equivalently write $p = \mathbb{E}\left[\mathbf{1}_{\{\lambda_1^{\varphi} > \mu\}}\right]$, where $\mathbf{1}_{\{\cdot\}}$ denotes the indicator function. Thus, we can write,

$$\begin{aligned}\frac{\partial p}{\partial {\lambda^A}_1} &= \frac{\partial p}{\partial \lambda_1^{\varphi}} \frac{\partial \lambda_1^{\varphi}}{\partial {\lambda^A}_1} \\ &= \mathbf{1}_{\{\lambda_1^{\varphi} > \mu\}} \left(2L-2\right) \eta_1^2 \left(1 + {\lambda^A}_1 {\lambda^H}_d\right)^{2L-3} \\ &\geq 0,\end{aligned}$$

where the last line follows since all terms on the right-hand side of the previous line are non-negative for $L > 1$. This completes the proof of Lemma 4.2. $\square$

It follows that,

$$\Delta p \times \Delta {\lambda^A}_1 \geq 0, \tag{13}$$

which establishes a positive correlation between p and ${\lambda^A}_1$. Meanwhile Bauer et al. (2011) show that ${\lambda^L}_n$ is bounded above by a term that is monotonic in the minimum ORC, where ${\lambda^L}_n$ denotes the maximum eigenvalue of the normalized Laplacian of A. Notice that A is row stochastic (Dovonon et al., 2024), which implies that its normalized Laplacian is in fact $I - A$. Therefore, ${\lambda^L}_n = 1 - {\lambda^A}_1$, and hence, the bound on the spectrum of the normalized Laplacian, which depends on a monotonic function of the minimum ORC, also bounds $1 - {\lambda^A}_1$ from above. Considering Equation 13, it follows immediately that a monotonic function of the minimum ORC controls a variable that is positively correlated with p, establishing the link between ORC and convergence of gradient descent. This suggests that modifying the ORC distribution of the attention map impacts transformers training.

Discussion on the assumptions. The self-attention mechanism is in fact a common setup for transformers, hence the only constraint assumption **A0** imposes beyond common practice is sharing the projection matrix between keys and queries. This yields a symmetric attention, while restricting the keys and queries projections to the column space of a shared weight matrix. Assumptions **A3** and **A5** are purely technical assumptions for our proof. Regarding Assumption **A4**, first note that the order in which the weight matrices are combined to parameterize F with θ can be completely flexible. Therefore, it suffices if assumption **A4** holds for any arrangement of the weights in θ. Despite this flexibility, being simultaneously diagonalizable restricts the degrees of freedom in M and $\nabla_\theta M$. This assumption implies that the two matrices share an eigenstructure, though they could have different eigenvalues. From a geometric perspective, this assumption limits the degrees of freedom that the changes in M have at each gradient step to the directions of its eigenvectors. While this is a restrictive assumption, we should keep in mind that transformers are often over-parameterized with a high degree of freedom, and hence the assumption is still plausible. Furthermore, we assume **A4** for technical reasons in order to simplify the proof of the theorem, and otherwise, the assumption is not a conceptual necessity. In other words, while assuming **A4** facilitates our proof, it is only a tool to simplify a step of the proof, not the only tool to show the main conclusion, and the implications may still hold without it.

B Graph geometry and statistical mechanics of transformers

In Section 3 we discussed the value of looking at transformers from the angle of networks/graphs and its consequences for training and robustness of transformers. The mathematical implications of this perspective for training transformers are provided in Section 4. Continuing in this direction from the perspective of network dynamics, in this appendix we elaborate on the physics that follow as an input signal is processed by a series of attention blocks (graph operators), which enables us to analyze the robustness properties of transformers from a networked system point of view. As we mentioned in Section 3, we interpret attention as a network of interacting particles and study the statistical mechanics of transformers in light of established results on the interplay between graph geometry and the dynamics and robustness of networked systems.

Attention as a particle system and transformer dynamics. A forward pass through attention blocks in transformers can be interpreted as dynamics of a system of interacting particles where each attention score represents the strength of the interaction between a pair of particles (tokens).